Examiner only

2. Radiotherapy is used to treat cancer. Three types of radiotherapy are described below.

Brachytherapy is a type of internal radiotherapy. It involves putting a sealed radiation source inside the cancerous growth. The radioisotope used emits low energy gamma rays. An isotope of iodine (iodine-125) can be used to treat prostate cancer.

Unsealed source radiotherapy also uses radioactive substances to treat cancer. These are introduced into the body by injection or ingestion. Iodine-131 is injected into a patient to treat thyroid cancer.

External radiotherapy is different from the methods described above. It is given as a series of short, daily treatments in the radiotherapy department using high energy gamma rays.

Information about some isotopes of iodine is given below.

Iodine-123 has a half-life of 13 hours and emits gamma.
Iodine-125 has a half-life of 59 days and emits gamma.
Iodine-128 has a half-life of 25 minutes and emits beta.
Iodine-129 has a half-life of 15 000 000 years and emits beta and gamma.
Iodine-131 has a half-life of 8 days and emits beta and gamma.

(a) Explain why iodine-123 is unsuitable to treat prostate cancer. [2]

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(b) Explain why iodine-131 is more suitable to treat thyroid cancer than iodine-128. [2]

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(c) Patients are told that, after treatment with iodine-131, small amounts of radiation from their body may trigger radiation monitors until the activity has dropped to one thousandth $\left(\frac{1}{1000}\right)$ of its initial value. The patients are told this will occur 80 days after treatment.
Explain with the aid of a calculation whether 80 days is long enough. [3]

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